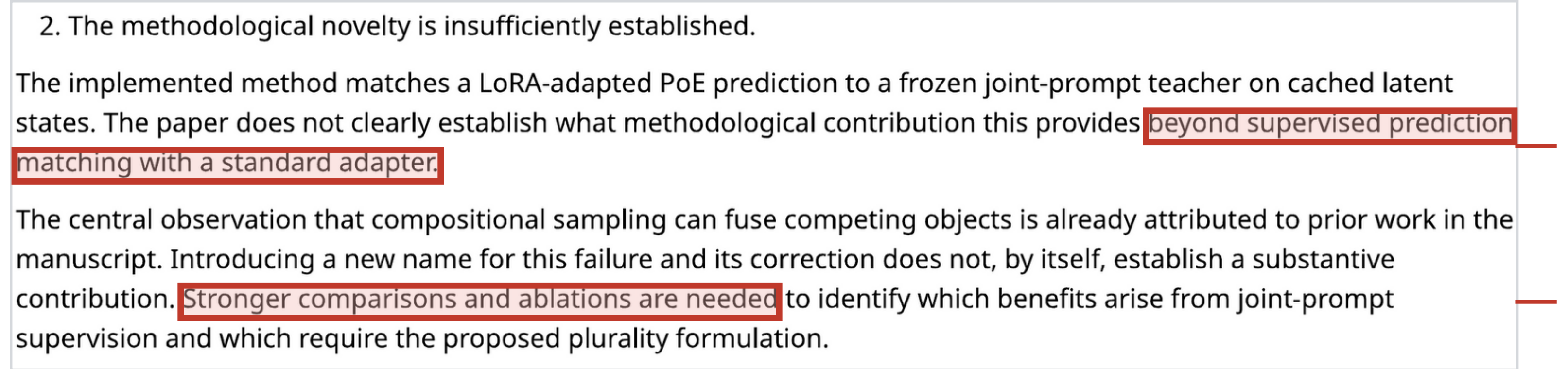
# G2: What is new in the method

k4nd weakness 2 · status: partly answered (the supervision ablation is future work)

**THE REVIEWER'S POINT**

### k4nd, weakness 2



2. The methodological novelty is insufficiently established.

The implemented method matches a LoRA-adapted PoE prediction to a frozen joint-prompt teacher on cached latent states. The paper does not clearly establish what methodological contribution this provides beyond supervised prediction matching with a standard adapter.

The central observation that compositional sampling can fuse competing objects is already attributed to prior work in the manuscript. Introducing a new name for this failure and its correction does not, by itself, establish a substantive contribution. Stronger comparisons and ablations are needed to identify which benefits arise from joint-prompt supervision and which require the proposed plurality formulation.

Reviews.pdf, page 4 (no text layer; passage located by OCR)

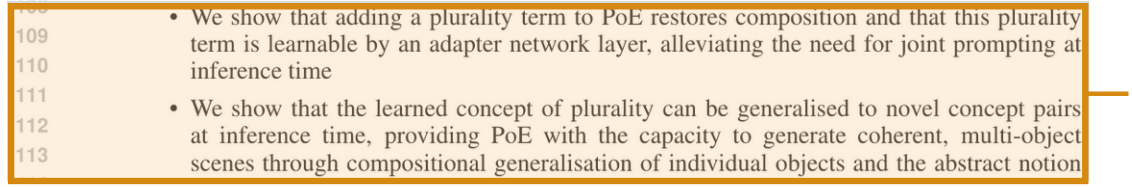
K-W2

K-W2

the novelty claimed

**THE SUBMITTED WEAKNESS**

### Introduction, the contributions



- We show that adding a plurality term to PoE restores composition and that this plurality term is learnable by an adapter network layer, alleviating the need for joint prompting at inference time
- We show that the learned concept of plurality can be generalised to novel concept pairs at inference time, providing PoE with the capacity to generate coherent, multi-object scenes through compositional generalisation of individual objects and the abstract notion

56173_Restoring_plurality_in_c.pdf (submitted), page 3

framed as a new concept

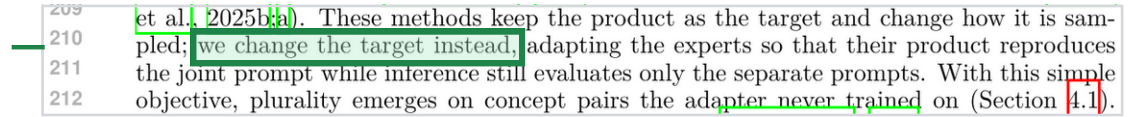
stated as

not run

**THE REVISED ANSWER**

### Section 2, prior corrections

what is new: the target



et al., 2025b;a). These methods keep the product as the target and change how it is sampled; we change the target instead, adapting the experts so that their product reproduces the joint prompt while inference still evaluates only the separate prompts. With this simple objective, plurality emerges on concept pairs the adapter never trained on (Section 4.1).

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 4

### Not in the revision

The ablation k4nd asks for (the same joint-prompt supervision on pairs the product does not fuse) was not run. The reply names it as future work.

no crop: this panel states what the claim table records

### The reply we plan to send (replies/k4nd-and-uhz6.md: k4nd W2)

partly answered

**W2: methodological novelty**

The method is deliberately simple: a standard adapter trained by prediction matching. Its contribution is the target. Existing corrections of product-of-experts sampling keep the product as the target and change how it is sampled (Du et al., 2023; Skreta et al., 2024, 2025; Dutta et al., 2026). We instead adapt the experts so that their product reproduces the joint prompt's prediction, and the joint prompt is never evaluated at inference. With this simple objective, plurality emerges on concept pairs the adapter never trained on (Section 4.1, Figure 5). Section 2 now states this distinction.

We agree that fusion under composition was identified before (Bradley et al., 2025), and we cite it as such. What we add is a measurement of the missing term during sampling: it is specific to the pair and seed, and concentrated in the first ten of 50 steps (Figures 2 to 4).

Table 1 compares the adapter with PoE and CO3 on the same seeds, and Figure 5 adds a Langevin corrector. The ablation the reviewer describes, the same joint-prompt supervision on pairs the product does not fuse, would separate the effect of the supervision from that of the plurality pairs; we have not run it and name it as future work.

Changes behind this page: Section 2 says prior corrections keep the product as the target while this paper changes the target.
Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.